

behave as separate stocks from the main stock. Local depletion of sprat in such areas can be an issue of ecological concern.

11.12 Ecosystem Considerations

Sprat is an important prey species in the North Sea ecosystem. The influence of the sprat fishery on other fish species and seabirds are at present not documented to be substantial.

In the North Sea, the key predators consuming sprats are included in the stock assessment, using SMS estimates of sprat consumption for each predatory fish stock, and estimates for seabirds though this information is as described under natural mortality not up to date. Impacts of changes in zooplankton communities and consequent changes in food densities for sprat are not included in the assessment, but it may be useful to explore the possibility of including this, or a similar proxy bottom-up driver, in future assessments. However, the effect of changes in productivity is included in the observed quarterly weight-at-age and in the estimated recruitment, as a decline in e.g., available food can lead to lower observed weights and lower estimated recruitment even in the absence of a causal link in the model.

11.13 Changes in the environment

Temperatures in this area have been increasing over the last few decades. This may have implications for sprat, although the correlation between temperature and recruitment from the model has been found to be low (see WKSPRAT: ICES, 2018a).

11.14 Tables and Figures

Table 11.1.1. North Sea & 3.a sprat. Landings (' 000 t) 2005–2024. See ICES (2006) for earlier data. Catch in coastal areas of Norway excluded. Data provided by Working Group members. These figures do not in all cases correspond to the official statistics and cannot be used for management purposes.

Country	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Division 27.4.a																				
Denmark		*	0.8	*	*					*	*	0.1	0.1		*	0.5	*	*		
Norway					*		*								0.1	*				
Sweden																				
UK (Scotland)							0.5						*	*						
Germany											*	*						*		
Netherlands											*									
France																			*	
Total		*	0.8	*	*		0.5			*	*	0.1	0.1	*	0.1	0.5	*	*		
Division 27.4.b																				
Denmark	204	79.5	55.5	51.4	115.6	80.8	90.9	65.7	44.7	121.3	234.4	177.6	100.6	156.5	110.3	138.4	66	79.7	76.4	71.2
Norway		0.8	3.7	1.3	4	8	0.1	6.2	*	8.9	0.3	19.6	9.7	9.3	10	9.3			3.6	0.7
Sweden	*				0.3	0.6	1.1	1.8	0.1	3.9	5.5	11.7	8.1	7.6	7.5	3.5	5.9	6.6	8.7	7.1
UK (Scotland)			0.1		2.5	1.1	1.9	0.7						*	1.3	1.7	*	0.4		
UK (Engl. & Wales)					*								*	*		0.1		0.2	*	

Country	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Germany							3.3	0.5	0.6	1.5	3.1	5.4	6	3.7	3.4	10	3.6	2.4		2.7
Netherlands							1.1	2.7	0.4	2.4	1.2	1	1.6	1.6		0.5		0.4	*	*
Faroe Islands												4.7	1	1		1				
Total	204.1	80.3	59.3	52.7	122.4	90.4	98.4	77.5	45.8	138	244.6	220	127	179.7	132.6	164.7	75.5	89.6	88.7	81.9
Division 27.4.c																				
Denmark	2	23.8	20.6	8.1	8.2	48.5	20	3.2	15.4	2.2	34	18.7	1.5	6.2	8.9	2.4	2.7		2	0.5
Norway		9	2.9		1.8	3.2	9.9	3	1.7	0.1	8.8	0.6		0.5	0.6	0.7				
Sweden					0.6	0.6	0.2	0.4	1.3		1.2	0.4					1.1		0.5	
UK (Scotland)				0.2			0.4					*				0.7	0.1			
UK (Engl. & Wales)	1.6	0.5	0.3	*	*	0.8	0.6	0.5	*	*	*	*	*	0.1	0.2	0.1	*	0.1	0.1	0.1
Germany							*	*	1		0.6	0.2				0.1				
Netherlands							4.2	1	0.7	*	1.2	0.8	*	0.7		1.6	0.1	*	*	*
Belgium							*		*	*	*	*	*		*	*	*	*	*	*
France											*		*							
Total	3.6	33.4	23.8	8.4	10.6	53	35.2	8	20.1	2.3	45.8	20.6	1.6	7.5	9.6	5.6	4	0.1	2.6	0.6
Division 27.3.a																				
Denmark	31.9	7.8	9.9	5.8	6.9	8.4	8	8.4	1.9	16.7	11.7	6.7	1	2.9	3.9	9.5	0.6	0.3	*	1.8
Sweden	7.7	4.4	4.2	2.4	1.6	1.4	2	1.5	1.1	1.5	1.3	1.1	0.2	1.1	1.7	2.4	0.7	0	0.1	0.7
Germany										*				*						
Faroe Islands												*								
Total	39.6	12.2	14.1	8.2	8.5	9.8	10	9.9	3	18.3	13	7.9	1.2	4	5.6	11.9	1.3	0.4	0.1	2.5
Total North Sea and Skagerrak-Kattegat																				
Denmark	237.9	111.2	86.7	65.4	130.7	137.7	119	77.4	62.1	140.2	280.1	203.1	103.3	165.6	123.1	150.9	69.3	80.1	78.4	73.6
Norway		9.8	6.7	1.3	5.8	11.1	10	9.1	1.7	9	9.1	20.2	9.7	9.8	10.6	10			3.6	0.7
Sweden	7.8	4.4	4.2	2.4	2.5	2.6	3.3	3.7	2.5	5.4	8.1	13.2	8.3	8.7	9.2	5.9	7.6	6.6	9.2	7.8
UK (Scotland)			0.1	0.2	2.5	1.1	2.8	0.7				*	*	*	1.3	2.5	0.1	0.4		
UK (Engl. & Wales)	1.6	0.5	0.3	*	*	0.8	0.6	0.5	*	*	*	*	*	*	0.2	0.2	*	0.3	0.1	0.1
Germany							3.3	0.5	1.6	1.6	3.7	5.6	6	3.7	3.4	10.1	3.6	2.4		2.7
Netherlands							5.3	3.7	1.1	2.4	2.4	1.8	1.6	2.3		2.1	0.1	0.4	*	*
Faroe Islands												4.7	1	1		1				
Belgium							*		*	*	*	*	*		*	*	*	*	*	*
France											*		*					*		
Total	247.3	125.9	97.9	69.3	141.6	153.3	144.1	95.5	68.9	158.7	303.3	248.5	129.9	191.2	147.8	182.7	80.8	90.1	91.4	85.0

Table 11.1.2. North Sea & 3.a sprat. Catches (tonnes) by quarter. Catches in coastal areas of Norway excluded. Data for 1996–1999 in ICES (2006).

Year	Quarter	27.4.a	27.4.b	27.4.c	27.3.a	Total
2000	1		18 126	28 063		46 189
	2		1722	45		1767
	3		131 306	1216		132 522
	4		12 680	2718		15 398
	Total		163 834	32 042		195 876
2001	1	115	40 903	9716		50 734
	2		1071			1071
	3		44 174	481		44 655
	4	79	65 102	8538		73 719
	Total	194	151 249	18 735		170 177
2002	1	1 136	2182	2790		6108
	2		435	93		528
	3		70 504	647		71 151
	4		52 942	12 911		65 853
	Total	1 136	126 063	16 441		143 640
2003	1		11 458	7727	5217	24 402
	2		625	26	1397	2049
	3		56 207	165	1720	58 092
	4		84 629	15 651	7349	107 629
	Total		152 919	23 570	15 683	192 172
2004	1		827	1831	4456	7113
	2	7	260	16	1510	1793
	3		54 161	496	4138	58 794
	4		120 685	15 937	10 775	147 397
	Total	7	175 932	18 280	20 879	215 097
2005	1		11 538	2457	8148	22 143
	2		2515	123	4722	7360
	3		107 530		19 418	126 948
	4		82 474	1033	7296	90 803
	Total		204 057	3613	39 584	247 254
2006	1	47	13 713	33 534	8105	55 399
	2		190	8	324	522
	3		40 051	8	1440	41 499
	4	2	26 579	77	2335	28 993
	Total	49	80 533	33 627	12 204	126 413
2007	1		582	247	2646	3475
	2		241	3	1291	1535
	3		16 603		5357	21 960
	4	769	41 850	23 531	4761	70 911
	Total	769	59 276	23 781	14 055	97 881
2008	1		2872	43	2890	5805
	2		52	*	1017	1069
	3		21 787		636	22 423

Year	Quarter	27.4.a	27.4.b	27.4.c	27.3.a	Total
2012	1		81	1649	4668	6399
	2		2924	0	909	3832
	3		26 779	307	1631	28 717
	4		47 765	6060	2728	56 553
	Total		77 549	8016	9936	95 501
2013	1		1281	3158	1296	5734
	2		32	0	443	474
	3		25 577	720	211	26 509
	4		18 892	16 276	943	36 110
	Total		45 781	20 154	2893	68 827
2014	1		59	125	384	568
	2		11 631	3	1415	13 050
	3	1	88 457	1428	9622	99 507
	4	7	37 851	822	6905	45 586
	Total	8	137 999	2378	18 327	158 711
2015	1	*	14 816	16 972	1442	33 230
	2		16 843	107	619	17 568
	3		124 512	335	6528	131 375
	4	25	88 395	28 375	4389	121 184
	Total	25	244 566	45 789	12 978	303 358
2016	1	68	18 487	5969	746	25 250
	2		8927	51	669	9 647
	3	*	158 522	111	4664	163 297
	4	2	34 070	14 466	1764	50 301
	Total	70	220 007	20 596	7843	248 516
2017	1	1	3432	1220	92	4 745
	2		1327	0	33	1 360
	3	*	92 885	217	227	93 329
	4	94	29 310	174	849	30 426
	Total	95	126 954	1611	1200	129 860
2018	1	*	8994	1628	168	10 790
	2		11 898	0	224	12 122
	3		112 361	1	1328	113 690
	4		46 411	5922	2249	54 582
	Total	*	179 664	7551	3969	191 184
2019	1		389	9 592	627	10 609
	2	2	3 606	11	379	3 999
	3	2	95 829	7	2 249	98 087
	4	49	32 750	3	2 296	35 098
	Total	53	132 574	9 614	5 551	147 793
2020	1	3	298	1 076	378	1 746
	2		19 430	*	173	19 603
	3	2	120 890	*	4 268	125 160

[illegible] $* < 0.5 \text{ t}$

** Until the 1st of March

Year	Quarter	27.4.a	27.4.b	27.4.c	27.3.a	Total
	4	520	24 049	4 489	7 087	36 145
	Total	526	164 667	5 566	11 896	182 654
2021	1	0	137	236	445	818
	2	*	326	1	11	338
	3	1	63 401	902	57	64 361
	4	1	11 601	2 850	791	15 244
	Total	2	75 464	3 989	1 305	80 761
2022	1		82	85	331	499
	2		19 449		16	19 465
	3	*	52 852			52 852
	4	8	17 237	8	36	17 289
	Total	8	89 620	94	383	90 105
2023	1		7	130	1	138
2023	2		479	0	34	512
2023	3		69 151	1	17	69 168
2023	4		19 062	2475	64	21 601
2023	Total		88 698	2606	116	91 420
2024	1		566	526	2	1 094
2024	2		6 635	10	107	6 752
2024	3		59 487	76	658	60 222
2024	4		15 173	*	1 729	16 901
2024	Total		81 862	613	2 496	84 970
2025	1**		2		1	2
2025	2					
2025	3					
2025	4					
2025	Total					

Table 11.2.1. North Sea & 3.a sprat. Species composition in Danish sprat fishery in tonnes and percentage of the total catch. Left: North Sea, right: Division 3.a.

Unit	Year	Area	Sprat	Herring	Horse mackerel	Whiting	Haddock	Mackerel	Cod	Sandeel	Other	Total
t	1998	27.4	129 315	11 817	573	673	6	220	11	2 174	1 187	145 978
t	1999	27.4	157 003	7 256	413	1 088	62	321	7	4 972	635	171 757
t	2000	27.4	188 463	11 662	3 239	2 107	66	766	4	423	1 911	208 641
t	2001	27.4	136 443	13 953	67	1 700	223	312	4	17 020	1 141	170 862
t	2002	27.4	140 568	16 644	2 078	2 537	27	715	0	4 102	801	167 471
t	2003	27.4	172 456	10 244	718	1 106	15	799	11	5 357	3 504	194 210
t	2004	27.4	179 944	10 144	474	334	0	4 351	3	3 836	1 821	200 906
t	2005	27.4	201 331	21 035	2 477	545	4	1 009	16	6 859	974	234 251
t	2006	27.4	103 236	8 983	577	343	25	905	4	5 384	576	120 033
t	2007	27.4	74 734	6 596	168	900	6	126	18	6	253	82 807
t	2008	27.4	61 093	7 928	26	380	10	367	0	23	1 735	71 563
t	2009	27.4	112 721	7 222	44	307	3	116	1	1 526	407	122 345
t	2010	27.4	112 395	4 410	11	119	2	18	0	1 236	577	118 769
t	2011	27.4	109 376	8 073	35	191	0	127	0	1 881	345	120 026
t	2012	27.4	67 263	8 573	2	354	0	246	0	93	411	76 943
t	2013	27.4	55 792	5 176	47	445	0	277	2	1	369	62 109
t	2014	27.4	123 180	11 402	0	897	0	70	16	16	1 700	137 280
t	2015	27.4	265 356	4 568	5	1 809	0	527	0	147	3 311	275 723
t	2016	27.4	192 718	11 107	18	4 223	0	439	0	46	2 093	210 643
t	2017	27.4	100 833	5 130	1	1 344	0	197	0	503	12 386	120 394
t	2018	27.4	161 536	7 528	174	716	0	366	0	24	344	170 687
t	2019	27.4	118 302	2 757	1	897	1	176	0	3	503	122 639
t	2020	27.4	140 954	6 227	19	898	93	1 188	0	11	724	150 114
t	2021	27.4	68 492	5 518	39	1 064	345	747	0	3	602	76 809
t	2022	27.4	78 825	3 829	1	488	124	397		4	227	83 895
t	2023	27.4	78 222	6 390	14	1 802	1 120	686		4	375	88 613
t	2024	27.4	71 202	3 881	12	1 414	916	497		8	211	78 140
%	1998	27.4	89	8	0	1	0	0	0	2	1	100
%	1999	27.4	91	4	0	1	0	0	0	3	0	100
%	2000	27.4	90	6	2	1	0	0	0	0	1	100
%	2001	27.4	80	8	0	1	0	0	0	10	1	100
%	2002	27.4	84	10	1	2	0	0	0	2	1	100
%	2003	27.4	89	5	0	1	0	0	0	3	2	100
%	2004	27.4	90	5	0	0	0	2	0	2	1	100
%	2005	27.4	86	9	1	0	0	0	0	3	0	100
%	2006	27.4	86	8	1	0	0	1	0	5	1	100
%	2007	27.4	90	8	0	1	0	0	0	0	0	100
%	2008	27.4	85	11	0	1	0	1	0	0	2	100
%	2009	27.4	92	6	0	0	0	0	0	1	0	100
%	2010	27.4	95	4	0	0	0	0	0	1	1	100
%	2011	27.4	91	7	0	0	0	0	0	2	0	100
%	2012	27.4	87	11	0	1	0	0	0	0	1	100
%	2013	27.4	90	8	0	1	0	0	0	0	1	100
%	2014	27.4	90	8	0	1	0	0	0	0	1	100
%	2015	27.4	96	2	0	1	0	0	0	0	1	100
%	2016	27.4	92	5	0	2	0	0	0	0	1	100
%	2017	27.4	84	4	0	1	0	0	0	0	10	100
%	2018	27.4	95	4	0	0	0	0	0	0	0	100
%	2019	27.4	97	2	0	1	0	0	0	0	0	100
%	2020	27.4	94	4	0	1	0	1	0	0	1	100
%	2021	27.4	90	6	0	1	1	1	0	0	1	100
%	2022	27.4	94	5	0	1	0	0	0	0	0	100
%	2023	27.4	88	7	0	2	1	1	0	0	0	100
%	2024	27.4	91	5	0	2	1	1	0	0	0	100
t	1998	27.3.a	9 143	3 385	230	467	54	0	49	7	2 866	16 202
t	1999	27.3.a	16 603	8 470	138	1 026	210	5	75	3 337	2 896	32 760
t	2000	27.3.a	12 578	8 034	5	1 062	308	8	52	13	3 556	25 617
t	2001	27.3.a	18 236	8 196	75	1 266	50	13	35	4 281	1 271	33 423
t	2002	27.3.a	11 451	12 982	21	1 164	3	6	30	606	2 280	28 541
t	2003	27.3.a	8 182	4 928	340	252	4	4	4	1	567	14 282
t	2004	27.3.a	13 374	4 620	97	976	18	24	27	116	2 155	21 408
t	2005	27.3.a	30 157	6 171	244	871	63	18	20	746	1 758	40 047
t	2006	27.3.a	6 814	2 852	215	276	13	3	45	1	232	10 451
t	2007	27.3.a	7 116	2 043	34	190	31	8	4	1	469	9 896
t	2008	27.3.a	4 805	1 948	14	285	0	0	11	462	39	7 563
t	2009	27.3.a	4 839	3 016	37	169	15	0	1	53	47	8 177
t	2010	27.3.a	2 851	2 134	25	142	6	1	2	135	171	5 466
t	2011	27.3.a	4 754	2 461	0	43	0	7	1	141	40	7 447
t	2012	27.3.a	5 707	5 495	9	149	7	10	5	0	228	11 610
t	2013	27.3.a	1 143	1 751	2	46	0	0	1	1	27	2 971
t	2014	27.3.a	16 751	3 777	5	343	1	20	5	12	888	21 801
t	2015	27.3.a	11 448	5 831	0	565	0	29	8	1	154	18 036
t	2016	27.3.a	7 001	2 140	0	335	1	19	3	0	78	9 579
t	2017	27.3.a	963	328	0	172	0	19	1	0	32	1 515
t	2018	27.3.a	2 872	257	2	150	1	11	0	0	12	3 304
t	2019	27.3.a	3 429	351	0	59	0	2	0	0	8	3 850
t	2020	27.3.a	9 494	551	4	249	5	41	1	0	27	10 372
t	2021	27.3.a	638	82	0	13	1	1	0	0	32	767
t	2022	27.3.a	316	19	0	1	0	0	0	0	31	368

Unit	Year	Area	Sprat	Herring	Horse mackerel	Whiting	Haddock	Mackerel	Cod	Sandeel	Other	Total
t	2023	27.3.a	17	4		0	0				0	21
t	2024	27.3.a	1 835	73	0	53	0	4	0	0	1	1 967
%	1998	27.3.a	56	21	1	3	0	0	0	0	18	100
%	1999	27.3.a	51	26	0	3	1	0	0	10	9	100
%	2000	27.3.a	49	31	0	4	1	0	0	0	14	100
%	2001	27.3.a	55	25	0	4	0	0	0	13	4	100
%	2002	27.3.a	40	46	0	4	0	0	0	2	8	100
%	2003	27.3.a	57	35	2	2	0	0	0	0	4	100
%	2004	27.3.a	63	22	1	5	0	0	0	1	10	100
%	2005	27.3.a	75	15	1	2	0	0	0	2	4	100
%	2006	27.3.a	65	27	2	3	0	0	0	0	2	100
%	2007	27.3.a	72	21	0	2	0	0	0	0	5	100
%	2008	27.3.a	64	26	0	4	0	0	0	6	1	100
%	2009	27.3.a	59	37	1	2	0	0	0	1	1	100
%	2010	27.3.a	52	39	1	3	0	0	0	3	3	100
%	2011	27.3.a	64	33	0	1	0	0	0	2	1	100
%	2012	27.3.a	49	47	0	1	0	0	0	0	2	100
%	2013	27.3.a	39	59	0	2	0	0	0	0	1	100
%	2014	27.3.a	77	17	0	2	0	0	0	0	4	100
%	2015	27.3.a	64	32	0	3	0	0	0	0	1	100
%	2016	27.3.a	73	22	0	4	0	0	0	0	1	100
%	2017	27.3.a	64	22	0	11	0	1	0	0	2	100
%	2018	27.3.a	87	8	0	5	0	0	0	0	0	100
%	2019	27.3.a	89	9	0	2	0	0	0	0	0	100
%	2020	27.3.a	92	5	0	2	0	0	0	0	0	100
%	2021	27.3.a	83	11	0	2	0	0	0	0	4	100
%	2022	27.3.a	86	5	0	0	0	0	0	0	8	100
%	2023	27.3.a	81	19	0	0	0	0	0	0	0	100
%	2024	27.3.a	93	4	0	3	0	0	0	0	0	100

Table 11.2.2. North Sea & 3.a sprat. Catch in numbers by age (1000's) by season and year. (Model year, e.g., 2021 = July 2021–June 2022)

Catch-at-age used as input for the assessment model (years refer to the model years)					
<i>Note that all catches in S4 have been moved to S1 in the following year and that all catches in S3 have been moved to S2 for the period 2019-2023</i>					
Year	Season	age 0	age 1	age 2	age 3
1974	1	0	16101061	2155723	475613
1974	2	1884146	11544114	866399	48228
1974	3	2842702	11091303	1336036	34534
1974	4	1302331	2511315	359117	14822
1975	1	250931	27723510	10052550	260182
1975	2	1179567	14541887	4378415	166807
1975	3	5240024	4755878	2206781	66186
1975	4	0	0	0	0
1976	1	2143211	42209830	2888653	180913
1976	2	7439656	18762732	1613139	88604
1976	3	7703416	6925346	267638	8289
1976	4	0	0	0	0
1977	1	2690194	12786056	5181867	109712
1977	2	2520082	4904593	3679153	67688
1977	3	15857197	1843468	2200876	37836
1977	4	0	0	0	0
1978	1	454090	32184524	427473	96435
1978	2	5517665	10344970	1209584	116695
1978	3	6154606	4973568	1119045	29941

Catch-at-age used as input for the assessment model (years refer to the model years)

Note that all catches in S4 have been moved to S1 in the following year and that all catches in S3 have been moved to S2 for the period 2019-2023

Year	Season	age 0	age 1	age 2	age 3
1978	4	0	0	0	0
1979	1	3579389	36866800	644042	117139
1979	2	1052920	11355949	2152261	63386
1979	3	3882781	6399259	332781	25964
1979	4	0	0	0	0
1980	1	0	14237558	17421360	1481066
1980	2	0	9415158	11520576	979415
1980	3	2536060	3866612	389674	8724
1980	4	0	0	0	0
1981	1	428776	12322431	1483241	130805
1981	2	40632	3540737	3025289	202048
1981	3	374254	3854059	319763	9835
1981	4	0	0	0	0
1982	1	545769	6350511	601581	64879
1982	2	818525	5021082	1070960	55333
1982	3	2530673	401839	46913	3525
1982	4	0	0	0	0
1983	1	5613728	2819244	969599	155653
1983	2	2375763	1334333	588678	91112
1983	3	1697718	596857	7271	0
1983	4	0	0	0	0
1984	1	954757	6475021	417235	2532
1984	2	521866	2535354	247654	4803
1984	3	405095	612407	10648	1053
1984	4	0	0	0	0
1985	1	0	1304457	1972027	37680
1985	2	0	576004	870780	16638
1985	3	84760	215856	150819	14916
1985	4	0	0	0	0
1986	1	0	177780	452745	347620
1986	2	0	156913	399604	306818
1986	3	580936	58710	740	0
1986	4	0	0	0	0
1987	1	2236	2250587	128512	2525
1987	2	49451	1790264	267597	978
1987	3	209788	826994	34626	32980
1987	4	0	0	0	0

Catch-at-age used as input for the assessment model (years refer to the model years)

Note that all catches in S4 have been moved to S1 in the following year and that all catches in S3 have been moved to S2 for the period 2019-2023

Year	Season	age 0	age 1	age 2	age 3
1988	1	4082942	2096911	2830054	42364
1988	2	1163964	314106	527986	11526
1988	3	1817700	637489	129384	5491
1988	4	0	0	0	0
1989	1	12451	1706824	3613841	5716
1989	2	783	76415	88925	342
1989	3	469458	416920	34789	12751
1989	4	0	0	0	0
1990	1	1568	2633068	2234213	342514
1990	2	1225	2058041	1746290	267714
1990	3	291837	62050	1941	429
1990	4	0	0	0	0
1991	1	40504	1684266	2416750	8159
1991	2	1552315	2936717	614233	9587
1991	3	208352	64565	1036	99
1991	4	0	0	0	0
1992	1	18948	9695465	1315325	177584
1992	2	222991	1185132	132166	16491
1992	3	1279875	1583952	259251	5821
1992	4	0	0	0	0
1993	1	264173	3026867	5339043	247839
1993	2	1441317	4911453	1324444	31435
1993	3	1867838	1819506	338969	43965
1993	4	0	0	0	0
1994	1	445326	40720484	516854	100737
1994	2	1856101	7146622	1455656	142774
1994	3	818875	2936362	559871	22813
1994	4	0	0	0	0
1995	1	170693	24466578	3192395	371759
1995	2	612010	8620522	2863267	505875
1995	3	1797666	4488224	533786	128194
1995	4	0	0	0	0
1996	1	299367	233497	816511	286503
1996	2	1083655	776795	2208631	911256
1996	3	1670742	289815	113580	49534
1996	4	0	0	0	0
1997	1	6447	2286585	130593	202822

Catch-at-age used as input for the assessment model (years refer to the model years)

Note that all catches in S4 have been moved to S1 in the following year and that all catches in S3 have been moved to S2 for the period 2019-2023

Year	Season	age 0	age 1	age 2	age 3
1997	2	148657	4395265	1078225	277615
1997	3	596223	728240	181187	46667
1997	4	0	0	0	0
1998	1	86124	3567341	1498339	258993
1998	2	5465889	2665032	1451844	326463
1998	3	1615982	1096547	489541	241493
1998	4	0	0	0	0
1999	1	830	15939248	477815	69219
1999	2	90557	2456063	254931	44836
1999	3	1967130	3351942	641059	183015
1999	4	0	0	0	0
2000	1	6101	9822669	1767256	70160
2000	2	81906	801375	384854	49827
2000	3	1093613	2807143	1310052	176418
2000	4	0	0	0	0
2001	1	13056	5767627	315550	7694
2001	2	550512	3967343	1528712	498496
2001	3	143017	531588	59709	13418
2001	4	0	0	0	0
2002	1	63416	6586442	594557	108679
2002	2	927294	4326530	661656	59022
2002	3	1182692	1199165	296900	65718
2002	4	0	0	0	0
2003	1	197639	4003316	594498	68144
2003	2	2785630	6826281	1115905	218400
2003	3	713229	39824	29774	26427
2003	4	0	0	0	0
2004	1	229309	4217281	731500	78913
2004	2	24806798	4735686	264373	53425
2004	3	5233945	309955	44145	15707
2004	4	0	0	0	0
2005	1	97602	13409729	479222	88858
2005	2	839944	7903545	228337	22051
2005	3	1089274	5408581	230703	38557
2005	4	0	0	0	0
2006	1	0	1987696	1401797	295158
2006	2	319709	493221	1003837	235542

Catch-at-age used as input for the assessment model (years refer to the model years)

Note that all catches in S4 have been moved to S1 in the following year and that all catches in S3 have been moved to S2 for the period 2019-2023

Year	Season	age 0	age 1	age 2	age 3
2006	3	176742	129541	176585	10933
2006	4	0	0	0	0
2007	1	0	1693273	189551	67672
2007	2	609939	4186796	1681648	254768
2007	3	404452	329724	19675	20964
2007	4	0	0	0	0
2008	1	11590	422430	1447939	329770
2008	2	2087187	1901763	1006626	260966
2008	3	893785	131774	41692	21858
2008	4	0	0	0	0
2009	1	0	4776947	219922	39037
2009	2	231412	8163927	554425	137328
2009	3	168362	3385107	519516	88967
2009	4	0	0	0	0
2010	1	12414	1732171	689166	90040
2010	2	349703	3105417	3011291	2157387
2010	3	298472	2412405	683264	90603
2010	4	0	0	0	0
2011	1	2469	1847215	1105017	281708
2011	2	420004	4234059	2917969	999295
2011	3	57320	250247	95834	42266
2011	4	0	0	0	0
2012	1	147896	2527701	729427	121665
2012	2	187098	3756225	1690250	281071
2012	3	78240	463743	86910	30157
2012	4	0	0	0	0
2013	1	10002	1973364	411558	72705
2013	2	462029	2176971	745578	144434
2013	3	193678	1554	2447	4794
2013	4	0	0	0	0
2014	1	2640874	9499013	627237	105519
2014	2	1215080	4046244	323320	92685
2014	3	1755944	2496884	177328	21685
2014	4	0	0	0	0
2015	1	1682642	12947813	2926867	161595
2015	2	615375	10862082	1632428	226924
2015	3	374504	1926029	733105	90223

Catch-at-age used as input for the assessment model (years refer to the model years)

Note that all catches in S4 have been moved to S1 in the following year and that all catches in S3 have been moved to S2 for the period 2019-2023

Year	Season	age 0	age 1	age 2	age 3
2015	4	0	0	0	0
2016	1	4450616	12775033	4537366	439570
2016	2	3593237	1451842	1251213	301252
2016	3	533954	47715	7358	2718
2016	4	0	0	0	0
2017	1	1767809	9076648	738627	88295
2017	2	1302514	2796713	182538	82806
2017	3	658881	807010	184005	68052
2017	4	0	0	0	0
2018	1	4548741	11562002	2878462	310552
2018	2	2090509	2888456	1516387	534059
2018	3	157673	1090798	254223	15776
2018	4	0	0	0	0
2019	1	2420231	9775216	3342785	163696
2019	2	1010279	2433674	1045309	140003
2019	3	0	0	0	0
2019	4	0	0	0	0
2020	1	207574	10153348	3429492	429318
2020	2	97488	2773937	394226	139520
2020	3	0	0	0	0
2020	4	0	0	0	0
2021	1	539434	5840604	1505982	255540
2021	2	286005	813983	392934	138813
2021	3	0	0	0	0
2021	4	0	0	0	0
2022	1	362805	7103177	813995	99384
2022	2	837407	285326	351175	101040
2022	3	0	0	0	0
2022	4	0	0	0	0
2023	1	578344	4941881	1983881	98132
2023	2	540868	1499195	609059	65056
2023	3	0	0	0	0
2023	4	0	0	0	0
2024	1	7184276	2925362	882840	85536
2024	2	2579168	537779	46599	8870
2024	3	0	0	0	0
2024	4	0	0	0	0

Table 11.2.3. North Sea & 3.a sprat. Mean weight at age (kg) in catches by season and year. (Model year, e.g., 2024 = July 2024–June 2025)

Weight-at-age used as input for the assessment model (years refer to the model years)					
<i>Note that weights in S4 are not used since there are no catches in S4</i>					
Year	Season	age 0	age 1	age 2	age 3
1974	1	0.0063	0.0083	0.0135	0.0184
1974	2	0.0058	0.0089	0.0150	0.0197
1974	3	0.0050	0.0077	0.0150	0.0197
1974	4	0.0066	0.0107	0.0183	0.0163
1975	1	0.0048	0.0086	0.0129	0.0172
1975	2	0.0075	0.0111	0.0168	0.0216
1975	3	0.0048	0.0106	0.0154	0.0192
1975	4	0.0062	0.0116	0.0170	0.0171
1976	1	0.0049	0.0070	0.0113	0.0134
1976	2	0.0043	0.0090	0.0153	0.0190
1976	3	0.0022	0.0059	0.0104	0.0126
1976	4	0.0034	0.0057	0.0085	0.0106
1977	1	0.0054	0.0082	0.0126	0.0180
1977	2	0.0059	0.0110	0.0146	0.0196
1977	3	0.0023	0.0080	0.0106	0.0138
1977	4	0.0025	0.0063	0.0083	0.0122
1978	1	0.0038	0.0069	0.0122	0.0146
1978	2	0.0044	0.0103	0.0155	0.0196
1978	3	0.0031	0.0089	0.0123	0.0166
1978	4	0.0020	0.0052	0.0087	0.0094
1979	1	0.0050	0.0058	0.0087	0.0113
1979	2	0.0057	0.0105	0.0150	0.0173
1979	3	0.0032	0.0077	0.0129	0.0165
1979	4	0.0029	0.0106	0.0121	0.0153
1980	1	0.0063	0.0052	0.0068	0.0083
1980	2	0.0051	0.0052	0.0069	0.0083
1980	3	0.0032	0.0086	0.0131	0.0168
1980	4	0.0046	0.0073	0.0105	0.0101
1981	1	0.0038	0.0099	0.0129	0.0156
1981	2	0.0082	0.0126	0.0153	0.0194
1981	3	0.0049	0.0089	0.0157	0.0194
1981	4	0.0060	0.0139	0.0191	0.0192
1982	1	0.0085	0.0089	0.0171	0.0155
1982	2	0.0071	0.0110	0.0160	0.0219
1982	3	0.0029	0.0075	0.0115	0.0174
1982	4	0.0044	0.0078	0.0114	0.0160
1983	1	0.0044	0.0092	0.0128	0.0152

Weight-at-age used as input for the assessment model (years refer to the model years)					
<i>Note that weights in S4 are not used since there are no catches in S4</i>					
Year	Season	age 0	age 1	age 2	age 3
1983	2	0.0042	0.0124	0.0169	0.0211
1983	3	0.0034	0.0094	0.0174	0.0163
1983	4	0.0038	0.0093	0.0127	0.0156
1984	1	0.0060	0.0081	0.0121	0.0166
1984	2	0.0053	0.0122	0.0168	0.0164
1984	3	0.0093	0.0135	0.0197	0.0197
1984	4	0.0093	0.0135	0.0197	0.0197
1985	1	0.0063	0.0093	0.0135	0.0197
1985	2	0.0051	0.0093	0.0135	0.0197
1985	3	0.0073	0.0099	0.0166	0.0166
1985	4	0.0073	0.0099	0.0166	0.0166
1986	1	0.0063	0.0073	0.0099	0.0166
1986	2	0.0051	0.0073	0.0099	0.0166
1986	3	0.0083	0.0164	0.0228	0.0163
1986	4	0.0084	0.0156	0.0208	0.0156
1987	1	0.0066	0.0086	0.0117	0.0153
1987	2	0.0060	0.0093	0.0112	0.0165
1987	3	0.0064	0.0125	0.0175	0.0206
1987	4	0.0068	0.0125	0.0167	0.0189
1988	1	0.0042	0.0088	0.0115	0.0138
1988	2	0.0046	0.0085	0.0113	0.0137
1988	3	0.0052	0.0132	0.0208	0.0158
1988	4	0.0063	0.0117	0.0155	0.0175
1989	1	0.0054	0.0086	0.0099	0.0170
1989	2	0.0044	0.0082	0.0109	0.0130
1989	3	0.0048	0.0077	0.0125	0.0155
1989	4	0.0046	0.0086	0.0115	0.0129
1990	1	0.0046	0.0070	0.0092	0.0115
1990	2	0.0038	0.0069	0.0092	0.0113
1990	3	0.0044	0.0099	0.0133	0.0156
1990	4	0.0048	0.0089	0.0119	0.0135
1991	1	0.0128	0.0143	0.0154	0.0168
1991	2	0.0048	0.0146	0.0189	0.0168
1991	3	0.0052	0.0101	0.0147	0.0172
1991	4	0.0062	0.0118	0.0152	0.0186
1992	1	0.0081	0.0099	0.0124	0.0148
1992	2	0.0058	0.0121	0.0153	0.0178
1992	3	0.0035	0.0096	0.0141	0.0179

Weight-at-age used as input for the assessment model (years refer to the model years)					
<i>Note that weights in S4 are not used since there are no catches in S4</i>					
Year	Season	age 0	age 1	age 2	age 3
1992	4	0.0042	0.0078	0.0104	0.0118
1993	1	0.0065	0.0109	0.0123	0.0138
1993	2	0.0075	0.0107	0.0135	0.0164
1993	3	0.0022	0.0080	0.0116	0.0152
1993	4	0.0023	0.0128	0.0154	0.0134
1994	1	0.0068	0.0067	0.0095	0.0129
1994	2	0.0087	0.0104	0.0125	0.0151
1994	3	0.0030	0.0082	0.0097	0.0140
1994	4	0.0038	0.0068	0.0090	0.0131
1995	1	0.0032	0.0082	0.0117	0.0121
1995	2	0.0051	0.0101	0.0133	0.0155
1995	3	0.0084	0.0096	0.0129	0.0158
1995	4	0.0058	0.0107	0.0142	0.0161
1996	1	0.0071	0.0108	0.0142	0.0175
1996	2	0.0079	0.0115	0.0150	0.0169
1996	3	0.0029	0.0062	0.0087	0.0103
1996	4	0.0031	0.0057	0.0077	0.0086
1997	1	0.0071	0.0128	0.0148	0.0163
1997	2	0.0058	0.0120	0.0161	0.0199
1997	3	0.0071	0.0097	0.0122	0.0147
1997	4	0.0052	0.0095	0.0127	0.0144
1998	1	0.0056	0.0139	0.0166	0.0186
1998	2	0.0050	0.0124	0.0153	0.0177
1998	3	0.0043	0.0061	0.0095	0.0094
1998	4	0.0039	0.0073	0.0097	0.0110
1999	1	0.0053	0.0097	0.0115	0.0121
1999	2	0.0046	0.0116	0.0135	0.0164
1999	3	0.0036	0.0094	0.0118	0.0138
1999	4	0.0052	0.0097	0.0129	0.0146
2000	1	0.0067	0.0122	0.0148	0.0185
2000	2	0.0062	0.0149	0.0174	0.0183
2000	3	0.0051	0.0105	0.0131	0.0150
2000	4	0.0036	0.0046	0.0080	0.0135
2001	1	0.0078	0.0109	0.0118	0.0159
2001	2	0.0048	0.0116	0.0136	0.0166
2001	3	0.0062	0.0127	0.0150	0.0162
2001	4	0.0065	0.0120	0.0161	0.0181
2002	1	0.0073	0.0109	0.0141	0.0154

Weight-at-age used as input for the assessment model (years refer to the model years)					
<i>Note that weights in S4 are not used since there are no catches in S4</i>					
Year	Season	age 0	age 1	age 2	age 3
2002	2	0.0077	0.0122	0.0142	0.0158
2002	3	0.0047	0.0101	0.0133	0.0145
2002	4	0.0060	0.0116	0.0129	0.0155
2003	1	0.0042	0.0125	0.0146	0.0228
2003	2	0.0058	0.0108	0.0145	0.0167
2003	3	0.0049	0.0115	0.0135	0.0141
2003	4	0.0050	0.0092	0.0123	0.0139
2004	1	0.0088	0.0116	0.0139	0.0154
2004	2	0.0041	0.0094	0.0126	0.0153
2004	3	0.0030	0.0097	0.0112	0.0130
2004	4	0.0044	0.0093	0.0115	0.0129
2005	1	0.0076	0.0097	0.0130	0.0154
2005	2	0.0066	0.0103	0.0115	0.0141
2005	3	0.0055	0.0080	0.0114	0.0138
2005	4	0.0047	0.0087	0.0115	0.0130
2006	1	0.0063	0.0108	0.0133	0.0152
2006	2	0.0055	0.0143	0.0158	0.0180
2006	3	0.0041	0.0095	0.0129	0.0134
2006	4	0.0050	0.0093	0.0124	0.0139
2007	1	0.0063	0.0119	0.0131	0.0149
2007	2	0.0065	0.0101	0.0127	0.0151
2007	3	0.0045	0.0075	0.0106	0.0126
2007	4	0.0048	0.0089	0.0118	0.0133
2008	1	0.0088	0.0103	0.0114	0.0131
2008	2	0.0044	0.0076	0.0126	0.0142
2008	3	0.0034	0.0076	0.0082	0.0085
2008	4	0.0044	0.0068	0.0090	0.0081
2009	1	0.0063	0.0096	0.0123	0.0142
2009	2	0.0046	0.0095	0.0130	0.0160
2009	3	0.0043	0.0077	0.0103	0.0135
2009	4	0.0087	0.0096	0.0105	0.0141
2010	1	0.0066	0.0080	0.0097	0.0137
2010	2	0.0047	0.0094	0.0114	0.0148
2010	3	0.0050	0.0072	0.0094	0.0130
2010	4	0.0038	0.0071	0.0095	0.0107
2011	1	0.0052	0.0085	0.0101	0.0134
2011	2	0.0044	0.0089	0.0114	0.0145
2011	3	0.0042	0.0102	0.0128	0.0171

Weight-at-age used as input for the assessment model (years refer to the model years)					
<i>Note that weights in S4 are not used since there are no catches in S4</i>					
Year	Season	age 0	age 1	age 2	age 3
2011	4	0.0050	0.0092	0.0123	0.0139
2012	1	0.0085	0.0087	0.0106	0.0150
2012	2	0.0072	0.0087	0.0119	0.0152
2012	3	0.0040	0.0069	0.0113	0.0146
2012	4	0.0047	0.0087	0.0117	0.0132
2013	1	0.0061	0.0096	0.0120	0.0150
2013	2	0.0043	0.0097	0.0124	0.0156
2013	3	0.0026	0.0051	0.0071	0.0084
2013	4	0.0022	0.0094	0.0128	0.0153
2014	1	0.0086	0.0086	0.0104	0.0168
2014	2	0.0070	0.0079	0.0116	0.0139
2014	3	0.0053	0.0083	0.0116	0.0119
2014	4	0.0065	0.0099	0.0101	0.0115
2015	1	0.0076	0.0082	0.0104	0.0150
2015	2	0.0072	0.0088	0.0109	0.0155
2015	3	0.0038	0.0078	0.0107	0.0153
2015	4	0.0044	0.0082	0.0109	0.0123
2016	1	0.0041	0.0077	0.0112	0.0145
2016	2	0.0051	0.0074	0.0118	0.0145
2016	3	0.0073	0.0143	0.0199	0.0235
2016	4	0.0076	0.0141	0.0188	0.0212
2017	1	0.0064	0.0083	0.0103	0.0139
2017	2	0.0038	0.0078	0.0099	0.0162
2017	3	0.0042	0.0064	0.0098	0.0130
2017	4	0.0076	0.0141	0.0188	0.0212
2018	1	0.0046	0.00664	0.0086	0.0126
2018	2	0.0053	0.0074	0.0097	0.0134
2018	3	0.0041	0.0067	0.0095	0.0136
2018	4	0.0057	0.0065	0.00762	0.0129
2019	1	0.0034	0.0064	0.0088	0.0116
2019	2	0.0041	0.0076	0.0098	0.0141
2019	3	0.0059	0.0099	0.0129	0.0163
2019	4	0.0064	0.0078	0.0105	0.0156
2020	1	0.0049	0.0093	0.0121	0.0162
2020	2	0.0071	0.0107	0.0141	0.0168
2020	3	0.0061	0.0087	0.0108	0.0138
2020	4	0.0065	0.0101	0.0131	0.0159
2021	1	0.0061	0.0071	0.0110	0.0131

Weight-at-age used as input for the assessment model (years refer to the model years)					
<i>Note that weights in S4 are not used since there are no catches in S4</i>					
Year	Season	age 0	age 1	age 2	age 3
2021	2	0.0061	0.0087	0.0117	0.0158
2021	3	0.0072	0.0122	0.0158	0.0200
2021	4	0.0070	0.0088	0.0103	0.0155
2022	1	0.0062	0.0084	0.0109	0.0135
2022	2	0.0076	0.0123	0.0166	0.0183
2022	3	0.0099	0.0167	0.0217	0.0274
2022	4	0.0065	0.0101	0.0131	0.0159
2023	1	0.0048	0.0091	0.0107	0.0125
2023	2	0.0051	0.0083	0.0108	0.0133
2023	3	0.0036	0.0060	0.0078	0.0098
2023	4	0.0047	0.0070	0.0099	0.0142
2024	1	0.0045	0.0082	0.0109	0.0136
2024	2	0.0046	0.0085	0.0123	0.0133
2024	3	0.0058	0.0100	0.0143	0.0166
2024	4	0.0065	0.0101	0.0131	0.0159

Table 11.2.4. North Sea and Division 3.a sprat. Sampling for biological parameters in 2024. This table only shows age-length samples, and therefore the number of samples may differ from Table 10.2.5.

Country	Quarter	Landings ('000 tonnes)	No. samples	No. measured	No. aged
Denmark	1	1.0	4	399	200
	2	5.5	4	458	195
	3	51.8	48	5011	2323
	4	15.3	26	3019	1223
	Total	73.6	82	8887	3941
Sweden	1	0.0	0	0	0
	2	0.0	2	100	100
	3	6.2	6	278	278
	4	1.6	6	298	298
	Total	7.8	14	676	676
All countries	1	1.0	4	3395	200
	2	5.5	6	558	295
	3	58.0	54	5289	2601

Country	Quarter	Landings (^{'000 tonnes})	No. samples	No. measured	No. aged
	4	16.9	32	3317	1521
Total	Total	81.4	96	9563	4671

Table 11.2.5. North Sea and Division 3.a sprat. Number of biological samples taken from 1974 and onward. The number of samples may differ from Table 10.2.4, since this table shows both length and age-length samples. These are the samples used to generate the catch-at-age matrix for the assessment model (Model year, e.g., 2024 = July 2024–June 2025).

Year	S1	S2	S3	S4
1974	15	31	102	25
1975	67	46	40	11
1976	54	70	53	16
1977	37	51	32	18
1978	52	78	47	22
1979	86	55	90	9
1980	0	0	49	28
1981	61	32	29	14
1982	27	48	13	16
1983	11	44	27	8
1984	9	23	29	7
1985	4	4	0	4
1986	4	1	0	1
1987	16	15	4	3
1988	8	4	9	1
1989	13	0	7	2
1990	4	0	13	1
1991	6	56	15	8
1992	42	35	24	4
1993	21	30	24	7
1994	42	50	32	5
1995	40	47	41	4
1996	2	12	8	3
1997	9	34	12	1
1998	25	38	16	3
1999	41	25	25	1
2000	29	23	22	14
2001	23	9	17	4
2002	26	37	28	7
2003	12	60	17	2
2004	26	43	24	15
2005	77	56	56	2

Year	S1	S2	S3	S4
2006	23	7	13	0
2007	34	40	13	4
2008	10	9	14	5
2009	33	36	18	5
2010	35	28	15	3
2011	28	57	20	3
2012	37	88	15	3
2013	31	23	2	10
2014	116	19	19	13
2015	165	47	21	2
2016	90	30	3	0
2017	69	21	11	6
2018	65	60	20	5
2019	65	45	2	12
2020	27	30	6	0
2021	85	22	0	8
2022	41	29	1	0
2023	78	49	4	6
2024	54	32	NA	NA

Table 11.3.1. North Sea sprat. Abundance indices by age from IBTS Q1

IBTS Q1 survey index (area 4 and 3a combined; years apply to the calendar year and ages the model year)				
<i>Index is calculated using a delta GAM model formulation (see Stock Annex)</i>				
Year	Age 0	Age 1	Age 2	Age 3
1983	252619	551262	574173	47111
1984	619180	553686	100186	25687
1985	374594	292408	75083	19254
1986	116338	137304	39250	9993
1987	503284	86061	25143	9769
1988	248663	789924	77117	15148
1989	744970	154929	114877	11326
1990	360108	185946	47580	21180
1991	1412224	176334	33438	7582
1992	1882139	281520	36961	9645
1993	1863182	1224852	103248	10709
1994	1195289	887347	132008	8288
1995	2258852	2257140	263386	10391
1996	604673	967027	199658	28253
1997	599335	270098	168138	27513
1998	1072937	1104108	180777	16056

IBTS Q1 survey index (area 4 and 3a combined; years apply to the calendar year and ages the model year)				
<i>Index is calculated using a delta GAM model formulation (see Stock Annex)</i>				
Year	Age 0	Age 1	Age 2	Age 3
1999	5183400	583736	73757	5308
2000	2017439	1164352	150449	25036
2001	1997862	1309083	239142	13995
2002	1191954	968965	87712	10393
2003	2493114	589410	66441	5540
2004	4084377	685280	106637	9076
2005	8918279	675529	29062	2718
2006	1230441	1416990	58676	7654
2007	1917763	1035569	162880	12506
2008	1526985	803061	47400	8526
2009	4133598	312030	34043	3833
2010	3288300	2489705	118665	17586
2011	1078333	926246	206207	47562
2012	3356603	3143308	245116	36666
2013	1137772	1116849	203191	29306
2014	3886605	443621	50655	9871
2015	7727188	3460669	317090	26651
2016	2112309	3409890	675849	37763
2017	10317128	1707447	128002	15146
2018	10440866	1547476	94598	11384
2019	6097175	2511994	226057	9585
2020	7316245	2219294	421523	40023
2021	3308192	1977916	196830	16693
2022	1810546	769303	57700	6537
2023	84401712	1710545	93914	7639
2024	1229364	336007	14974	3206
	7901271	2000434	138604	8294

Table 11.3.1. North Sea sprat. Abundance indices by age from IBTS Q3

IBTS Q3 survey index (area 4 and 3a combined; years and ages apply to both the model year and calendar year)			
<i>Index is calculated using a delta GAM model formulation (see Stock Annex)</i>			
Year	Age 1	Age 2	Age 3
1992	14555861	2633020	104865
1993	5767651	3015219	217792
1994	16468664	1326478	95089
1995	30622687	7433288	454582
1996	2317117	2219591	215543
1997	13080865	1171944	200385

IBTS Q3 survey index (area 4 and 3a combined; years and ages apply to both the model year and calendar year)			
<i>Index is calculated using a delta GAM model formulation (see Stock Annex)</i>			
Year	Age 1	Age 2	Age 3
1998	2676263	1107920	117795
1999	13792780	1719505	82599
2000	8212868	3228536	133847
2001	8998081	2277278	187452
2002	10011480	1319291	102476
2003	11610320	1272970	66231
2004	14371331	1945227	122791
2005	52835449	2266372	102272
2006	9340785	5459057	155440
2007	10549586	1552282	184767
2008	7894186	2085499	130785
2009	35252950	3032568	337850
2010	35355908	9422666	428224
2011	16742275	8341042	1191533
2012	11469646	5231406	575643
2013	9052264	3060010	414534
2014	63182232	3573736	215965
2015	59775893	18619852	653613
2016	27891385	4266699	482295
2017	27754797	2886164	173266
2018	18709889	3123833	200733
2019	40210818	8468920	521293
2020	53930015	16906066	1479519
2021	21858420	5602150	519985
2022	29786037	3579909	464099
2023	58581633	8893743	524218
2024	12220220	1048774	54697

Table 11.3.2. North Sea and Division 3.a sprat. HERAS survey index.

HERAS abundance index (area 4 and 3.a summed), data are from WGIPS (2019)			
<i>Years and ages apply to both the model year and calendar year</i>			
Year	Age 1	Age 2	Age 3
2006	21923	21368	1413
2007	42862	5837	2252
2008	17188	7868	840
2009	47690	16920	2815
2010	20328	14087	1174
2011	26581	14207	3412
2012	22036	12831	4693

HERAS abundance index (area 4 and 3.a summed), data are from WGIPS (2019)			
<i>Years and ages apply to both the model year and calendar year</i>			
Year	Age 1	Age 2	Age 3
2013	9347	6342	2049
2014	59020	20274	3982
2015	27082	22676	10142
2016	58604	33989	8160
2017	38135	3664	1465
2018	109180	10113	779
2019	93775	28020	5275
2020	38415	17993	2055
2021	46918	7051	1509
2022	60224	16200	2882
2023	48125	18542	7015
2024	30080	7800	695

Table 11.6.1. North Sea and Division 3.a sprat. Natural mortality input (Model year, e.g., 2024 = July 2024–June 2025). From multispecies SMS (WKSAM: ICES, 2023) 2022 key run.

Year	Season	age 0	age 1	age 2	age 3
1974	1	0.4533	0.4350	0.4200	0.3973
1974	2	0.3640	0.2157	0.1903	0.1630
1974	3	0.2493	0.2153	0.1680	0.1680
1974	4	0.3283	0.2923	0.2640	0.2640
1975	1	0.4533	0.4350	0.4200	0.3973
1975	2	0.3640	0.2157	0.1903	0.1630
1975	3	0.2633	0.2287	0.1723	0.1723
1975	4	0.3433	0.3227	0.2987	0.2987
1976	1	0.4250	0.4077	0.3917	0.3770
1976	2	0.3720	0.2280	0.2087	0.1800
1976	3	0.2890	0.2503	0.1907	0.1907
1976	4	0.3430	0.3237	0.2983	0.2983
1977	1	0.3687	0.3513	0.3290	0.3207
1977	2	0.3793	0.2310	0.2117	0.1833
1977	3	0.2940	0.2797	0.2190	0.2190
1977	4	0.3340	0.3170	0.2993	0.2993
1978	1	0.3230	0.3330	0.3097	0.2977
1978	2	0.3453	0.2137	0.2053	0.1807
1978	3	0.2837	0.2703	0.2123	0.2123
1978	4	0.3037	0.2780	0.2603	0.2603
1979	1	0.3453	0.3650	0.3357	0.3270
1979	2	0.3680	0.2767	0.2370	0.2163

Year	Season	age 0	age 1	age 2	age 3
1979	3	0.2973	0.2773	0.2257	0.2257
1979	4	0.3087	0.2917	0.2737	0.2737
1980	1	0.3733	0.3867	0.3647	0.3560
1980	2	0.3940	0.2900	0.2577	0.2227
1980	3	0.2917	0.2703	0.2043	0.2043
1980	4	0.3083	0.2890	0.2537	0.2537
1981	1	0.4007	0.4050	0.3843	0.3553
1981	2	0.3637	0.2963	0.2647	0.2180
1981	3	0.2770	0.2457	0.1887	0.1887
1981	4	0.2910	0.2697	0.2213	0.2213
1982	1	0.4017	0.3883	0.3760	0.3267
1982	2	0.3013	0.2227	0.2097	0.1627
1982	3	0.2237	0.1990	0.1533	0.1533
1982	4	0.2537	0.2203	0.1773	0.1773
1983	1	0.4143	0.3983	0.3880	0.3333
1983	2	0.2967	0.2237	0.1980	0.1680
1983	3	0.2137	0.1773	0.1357	0.1357
1983	4	0.2337	0.1990	0.1543	0.1543
1984	1	0.4243	0.4027	0.3990	0.3420
1984	2	0.3330	0.2293	0.2007	0.1730
1984	3	0.2283	0.1997	0.1393	0.1393
1984	4	0.2410	0.2097	0.1713	0.1713
1985	1	0.4513	0.4367	0.4303	0.3880
1985	2	0.4050	0.3040	0.2513	0.2143
1985	3	0.2423	0.1973	0.1390	0.1390
1985	4	0.2480	0.2123	0.1513	0.1513
1986	1	0.4777	0.4700	0.4583	0.4123
1986	2	0.4310	0.3310	0.2870	0.2357
1986	3	0.2627	0.2073	0.1483	0.1483
1986	4	0.2687	0.2230	0.1483	0.1483
1987	1	0.5007	0.4917	0.4810	0.4263
1987	2	0.4617	0.3457	0.3047	0.2507
1987	3	0.2730	0.1927	0.1653	0.1653
1987	4	0.2767	0.2223	0.1610	0.1610
1988	1	0.5227	0.5043	0.4927	0.4167
1988	2	0.4687	0.3283	0.3137	0.2753
1988	3	0.2863	0.2073	0.1850	0.1850
1988	4	0.2897	0.2340	0.1930	0.1930
1989	1	0.5177	0.5013	0.4883	0.4113
1989	2	0.4537	0.3327	0.3097	0.2857

Year	Season	age 0	age 1	age 2	age 3
1989	3	0.2857	0.2253	0.2017	0.2017
1989	4	0.2963	0.2530	0.2340	0.2340
1990	1	0.4920	0.4827	0.4627	0.4187
1990	2	0.4417	0.3290	0.2940	0.2850
1990	3	0.2730	0.2430	0.2133	0.2133
1990	4	0.2993	0.2617	0.2380	0.2380
1991	1	0.4580	0.4530	0.4373	0.4177
1991	2	0.4060	0.2917	0.2673	0.2480
1991	3	0.2627	0.2527	0.2080	0.2080
1991	4	0.3013	0.2717	0.2473	0.2473
1992	1	0.4283	0.4173	0.4087	0.3990
1992	2	0.3677	0.2557	0.2430	0.2130
1992	3	0.2527	0.2437	0.1953	0.1953
1992	4	0.2853	0.2560	0.2350	0.2350
1993	1	0.4133	0.4103	0.4007	0.3887
1993	2	0.2997	0.2310	0.2297	0.2053
1993	3	0.2417	0.2300	0.1833	0.1833
1993	4	0.2707	0.2420	0.2197	0.2197
1994	1	0.4487	0.4233	0.4057	0.4013
1994	2	0.3127	0.2430	0.2330	0.2193
1994	3	0.2363	0.2187	0.1833	0.1833
1994	4	0.2613	0.2243	0.2063	0.2063
1995	1	0.4543	0.4277	0.4100	0.3953
1995	2	0.3160	0.2433	0.2330	0.2193
1995	3	0.2107	0.1980	0.1727	0.1727
1995	4	0.2460	0.2017	0.1793	0.1793
1996	1	0.4607	0.4160	0.3983	0.3670
1996	2	0.3893	0.2580	0.2470	0.2170
1996	3	0.2107	0.1827	0.1527	0.1527
1996	4	0.2380	0.1867	0.1520	0.1520
1997	1	0.4237	0.3853	0.3690	0.3377
1997	2	0.3780	0.2380	0.2357	0.1880
1997	3	0.2070	0.1887	0.1473	0.1473
1997	4	0.2450	0.2027	0.1623	0.1623
1998	1	0.4117	0.3757	0.3533	0.3323
1998	2	0.4030	0.2313	0.2290	0.1907
1998	3	0.2237	0.2000	0.1620	0.1620
1998	4	0.2563	0.2210	0.1810	0.1810
1999	1	0.4050	0.3680	0.3537	0.3467
1999	2	0.3393	0.2263	0.2253	0.2037

Year	Season	age 0	age 1	age 2	age 3
1999	3	0.2323	0.2267	0.1927	0.1927
1999	4	0.2757	0.2447	0.2117	0.2117
2000	1	0.3900	0.3643	0.3583	0.3447
2000	2	0.3527	0.2370	0.2350	0.2223
2000	3	0.2157	0.2007	0.1840	0.1840
2000	4	0.2437	0.2080	0.1710	0.1710
2001	1	0.3923	0.3607	0.3600	0.3357
2001	2	0.3380	0.2507	0.2497	0.2383
2001	3	0.2200	0.2113	0.1897	0.1897
2001	4	0.2617	0.2263	0.1873	0.1873
2002	1	0.4010	0.3677	0.3670	0.3197
2002	2	0.3573	0.2537	0.2517	0.2417
2002	3	0.2100	0.1953	0.1683	0.1683
2002	4	0.2540	0.2123	0.1833	0.1833
2003	1	0.3867	0.3543	0.3537	0.3130
2003	2	0.3300	0.2520	0.2397	0.2397
2003	3	0.2307	0.2133	0.1907	0.1907
2003	4	0.3020	0.2590	0.2350	0.2350
2004	1	0.3910	0.3697	0.3607	0.3307
2004	2	0.3063	0.2393	0.2257	0.2257
2004	3	0.2140	0.1970	0.1840	0.1840
2004	4	0.2950	0.2523	0.2310	0.2310
2005	1	0.4023	0.3840	0.3747	0.3747
2005	2	0.3130	0.2490	0.2247	0.2247
2005	3	0.2067	0.1830	0.1693	0.1693
2005	4	0.2970	0.2493	0.2133	0.2133
2006	1	0.4503	0.4310	0.4217	0.4217
2006	2	0.3223	0.2417	0.2273	0.2250
2006	3	0.1927	0.1813	0.1653	0.1653
2006	4	0.2820	0.2410	0.2030	0.2030
2007	1	0.4697	0.4453	0.4417	0.4417
2007	2	0.3397	0.2560	0.2333	0.2310
2007	3	0.1867	0.1617	0.1440	0.1440
2007	4	0.2640	0.2213	0.1830	0.1830
2008	1	0.4720	0.4573	0.4473	0.4473
2008	2	0.3217	0.2377	0.2223	0.2077
2008	3	0.1803	0.1693	0.1583	0.1583
2008	4	0.2513	0.2237	0.1957	0.1957
2009	1	0.4547	0.4497	0.4350	0.4300
2009	2	0.3237	0.2507	0.2223	0.2073

Year	Season	age 0	age 1	age 2	age 3
2009	3	0.1807	0.1700	0.1593	0.1593
2009	4	0.2460	0.2187	0.1943	0.1943
2010	1	0.4997	0.4903	0.4680	0.4623
2010	2	0.3517	0.2760	0.2407	0.2257
2010	3	0.1970	0.1883	0.1630	0.1630
2010	4	0.2667	0.2237	0.2030	0.2030
2011	1	0.5360	0.5267	0.5007	0.4947
2011	2	0.3690	0.3050	0.2590	0.2520
2011	3	0.2113	0.2013	0.1767	0.1767
2011	4	0.2937	0.2517	0.2320	0.2320
2012	1	0.5507	0.5320	0.5067	0.4797
2012	2	0.3617	0.3000	0.2603	0.2507
2012	3	0.2017	0.1837	0.1457	0.1457
2012	4	0.2783	0.2273	0.1970	0.1970
2013	1	0.4817	0.4723	0.4517	0.4023
2013	2	0.3110	0.2633	0.2313	0.2217
2013	3	0.1847	0.1787	0.1527	0.1527
2013	4	0.2580	0.2270	0.2030	0.2030
2014	1	0.4363	0.4270	0.4077	0.3460
2014	2	0.3030	0.2487	0.2200	0.2133
2014	3	0.2013	0.1823	0.1547	0.1547
2014	4	0.2580	0.2283	0.2043	0.2043
2015	1	0.4373	0.4303	0.4060	0.3703
2015	2	0.3090	0.2713	0.2177	0.2163
2015	3	0.2217	0.2010	0.1750	0.1750
2015	4	0.2653	0.2450	0.2320	0.2320
2016	1	0.4573	0.4423	0.4140	0.4013
2016	2	0.3363	0.3033	0.2290	0.2277
2016	3	0.2460	0.2163	0.1927	0.1927
2016	4	0.2833	0.2643	0.2447	0.2447
2017	1	0.4567	0.4353	0.4080	0.4080
2017	2	0.3363	0.3140	0.2277	0.2277
2017	3	0.2387	0.2153	0.1937	0.1937
2017	4	0.2753	0.2577	0.2390	0.2390
2018	1	0.4257	0.4047	0.3787	0.3787
2018	2	0.3300	0.3080	0.2190	0.2190
2018	3	0.2437	0.2230	0.2120	0.2120
2018	4	0.2880	0.2720	0.2530	0.2530
2019	1	0.4243	0.4037	0.3800	0.3800
2019	2	0.3447	0.3227	0.2280	0.2280

Year	Season	age 0	age 1	age 2	age 3
2019	3	0.2670	0.2493	0.2390	0.2390
2019	4	0.3107	0.2943	0.2750	0.2750
2020	1	0.4447	0.4240	0.4007	0.4007
2020	2	0.3707	0.3490	0.2600	0.2600
2020	3	0.2953	0.2810	0.2717	0.2717
2020	4	0.3353	0.3190	0.2993	0.2993
2021	1	0.4453	0.4267	0.4050	0.4050
2021	2	0.3713	0.3527	0.2760	0.2760
2021	3	0.2953	0.2810	0.2717	0.2717
2021	4	0.3353	0.3190	0.2993	0.2993
2022	1	0.4453	0.4267	0.4050	0.4050
2022	2	0.3713	0.3527	0.2760	0.2760
2022	3	0.2953	0.2810	0.2717	0.2717
2022	4	0.3353	0.3190	0.2993	0.2993
2023	1	0.4453	0.4267	0.4050	0.4050
2023	2	0.3713	0.3527	0.2760	0.2760
2023	3	0.2953	0.2810	0.2717	0.2717
2023	4	0.3353	0.3190	0.2993	0.2993
2024	1	0.4453	0.4267	0.4050	0.4050
2024	2	0.3713	0.3527	0.2760	0.2760
2024	3	0.2953	0.2810	0.2717	0.2717
2024	4	0.3353	0.3190	0.2993	0.2993

Table 11.6.2. North Sea sprat. Assessment diagnostics.

Date: 03/18/25 Start time:14:13:20 run time:17 seconds

objective function (negative log likelihood): 289.662

Number of parameters: 149

Maximum gradient: 2.92474e-05

Akaike information criterion (AIC): 877.325

Number of observations used in the likelihood:

Catch	CPUE	S/R	Stomach	Sum
816	328	51	0	1195

objective function weight:

Catch	CPUE	S/R
1.00	1.00	0.10

unweighted objective function contributions (total):

Catch	CPUE	S/R	Stom.	Stom N.	Penalty	Sum
385.9	-97.1	8.6	0.0	0.0	0.00	297

unweighted objective function contributions (per observation):

Catch	CPUE	S/R	Stomachs
0.47	-0.30	0.17	0.00

contribution by fleet:

IBTS Q1 Rec	total: -14.594	mean: -0.339
IBTS Q1	total: -35.519	mean: -0.275
IBTS Q3	total: -30.869	mean: -0.312
Acoustic	total: -16.103	mean: -0.283

F, Year effect:

1974:	1.000
1975:	1.787
1976:	1.845
1977:	1.742
1978:	1.180
1979:	0.729
1980:	2.385
1981:	1.106
1982:	0.933
1983:	1.636
1984:	0.829
1985:	1.231
1986:	1.020
1987:	0.328
1988:	1.183

1989: 0.297
1990: 1.432
1991: 0.604
1992: 0.772
1993: 1.412
1994: 0.685
1995: 1.337
1996: 1.144
1997: 0.806
1998: 1.469
1999: 0.764
2000: 1.334
2001: 1.310
2002: 1.475
2003: 1.149
2004: 1.758
2005: 1.169
2006: 1.415
2007: 1.313
2008: 1.039
2009: 0.640
2010: 0.740
2011: 0.625
2012: 1.174
2013: 0.672
2014: 0.502
2015: 1.017
2016: 1.961
2017: 1.040
2018: 0.999
2019: 1.015
2020: 1.686
2021: 1.528
2022: 0.626
2023: 2.858
2024: 0.641

F, season effect:

age: 0

1974-2024: 0.041 0.206 0.415 0.250

age: 1

1974-2024: 0.586 0.540 0.250 0.250

age: 2

1974-2024: 0.285 0.512 0.158 0.250

age: 3

1974-2024: 0.257 0.553 0.302 0.250

F, age effect:

0 1 2 3

1974-2024: 0.047 0.448 1.371 1.371

Exploitation pattern (scaled to mean F=1)

0 1 2 3

1974-2024 season 1: 0.002 0.221 0.328 0.296

season 2: 0.008 0.203 0.589 0.637

season 3: 0.016 0.094 0.182 0.348

season 4: 0.010 0.094 0.288 0.288

sqrt(catch variance) ~ CV:

season

age 1 2 3 4

0 1.414 1.414 1.094 0.100

1 0.798 0.832 1.370 0.100

2 0.963 1.078 1.414 0.100

3 0.963 1.078 1.414 0.100

Survey catchability:

	age 0	age 1	age 2	age 3
IBTS Q1 Rec	598.350			
IBTS Q1	1.477	2.240	3.109	
IBTS Q3	0.823	1.081	0.883	
Acoustic	1.093	2.413	5.144	

Stock size dependent catchability (power model)

	age 0	age 1	age 2	age 3
IBTS Q1 Rec	1.52			
IBTS Q1	1.00	1.00	1.00	
IBTS Q3	1.00	1.00	1.00	
Acoustic	1.00	1.00	1.00	

sqrt(Survey variance) ~ CV:

	age 0	age 1	age 2	age 3
IBTS Q1 Rec	0.43			
IBTS Q1	0.46	0.46	0.46	
IBTS Q3	0.49	0.42	0.42	
Acoustic	0.40	0.49	0.49	

Average F:

sp. 1
1974: 1.152
1975: 1.771
1976: 1.856
1977: 1.807
1978: 1.231
1979: 0.772
1980: 2.396
1981: 1.115
1982: 0.917

1983: 1.563
1984: 0.806
1985: 1.178
1986: 0.975
1987: 0.317
1988: 1.136
1989: 0.295
1990: 1.410
1991: 0.611
1992: 0.778
1993: 1.394
1994: 0.680
1995: 1.293
1996: 1.102
1997: 0.790
1998: 1.446
1999: 0.777
2000: 1.317
2001: 1.306
2002: 1.452
2003: 1.168
2004: 1.749
2005: 1.161
2006: 1.371
2007: 1.252
2008: 0.997
2009: 0.621
2010: 0.714
2011: 0.607
2012: 1.106
2013: 0.650
2014: 0.495
2015: 1.007
2016: 1.925
2017: 1.039
2018: 1.018
2019: 0.921

2020: 1.523
2021: 1.387
2022: 0.586
2023: 2.511
2024: 0.599

Recruit-SSB		alfa	beta	recruit s2	recruit s
Sprat	Hockey stick -break.:	1288.351	9.000e+04	0.515	0.718

Table 11.6.3. North Sea and Division 3.a Sprat. Assessment output: Stock numbers (thousands) (years, seasons (S1-S4), and age (A0-A3+) refer to the model year, e.g., 2021 = July 2021–June 2022)

Year/Age Quarter	A0_S1	A0_S2	A0_S3	A0_S4	A1_S1	A1_S2	A1_S3	A1_S4	A2_S1	A2_S2	A2_S3	A2_S4	A3+_S1	A3+_S2	A3+_S3	A3+_S4
1974	395307000	25074100 0	17257700 0	131919000	120033000	59743900	37801500	2724560 0	1119830 0	4978830	2041320	1390420	551857	26095 0	103851	57997
1975	515815000	32668800 0	22316100 0	165674000	93898000	38005000	19875000	1294110 0	1818300 0	5944570	1403340	802936	789632	28310 3	62021	24884
1976	231803000	15101300 0	10227800 0	73923700	117529000	48147100	24522800	1552380 0	9372170	3081350	685683	380388	614082	22014 2	45370	17448
1977	379838000	26184600 0	17622200 0	126987000	52459100	23363800	12165900	7566240	1123130 0	4093920	976795	538679	295215	11612 3	25805	10070
1978	626428000	45249800 0	31676300 0	233152000	90929800	47808100	29020200	1940230 0	5510730	2550530	908210	569270	406794	19950 6	68072	33757
1979	351927000	24881200 0	17100900 0	125247000	172091000	98638700	62695800	4378110 0	1469330 0	7900520	3738070	2548250	464810	25936 2	120169	70876
1980	206302000	14138100 0	93188000	66477100	91984900	33394800	14028400	8193420	3270520 0	8947180	1298350	632308	1992070	60321 4	79110	23992
1981	61178700	40895600	28128200	20871900	48838800	24363900	13862300	9578760	6136970	2713430	959133	625520	509256	24198 6	84140	44058
1982	37138900	24809300	18191400	14285500	15602100	8280430	5287370	3902730	7314670	3488130	1469920	1030760	536632	27879 4	116748	68018
1983	50514300	33274800	24348500	19052300	11084800	4842840	2606130	1816870	3130970	1121230	292071	179111	920224	37094 9	90692	40190
1984	24950900	16297200	11589000	9076570	15082300	8109380	5274980	3936670	1489010	722709	330577	240425	187938	99737	44732	27594
1985	16828500	10690900	7046980	5400390	7132750	3335350	1826770	1306250	3192060	1283310	420912	280757	225817	99363	31522	16465
1986	60994900	37757300	24298500	18320900	4214240	2015000	1130680	819685	1056360	448552	164680	113915	255480	11819 0	43094	24346
1987	35362100	21420500	13457600	10177700	14004500	7857900	5136840	4083620	655843	356658	208936	164981	119200	69346	42082	31133

Year/Age Quarter	A0_S1	A0_S2	A0_S3	A0_S4	A1_S1	A1_S2	A1_S3	A1_S4	A2_S1	A2_S2	A2_S3	A2_S4	A3+_S1	A3+_S2	A3+_S3	A3+_S4
1988	50723500	30008100	18568700	13630200	7717830	3416240	1847740	1315260	3269540	1258740	401379	258395	166951	72618	22490	11448
1989	42693300	25427000	16107800	12036000	10202400	5716670	3814990	2945730	1040850	568862	339004	259899	222480	13285 1	79732	57631
1990	61190700	37310100	23662400	17517900	8949280	3791750	1929590	1288900	2287260	823292	224838	133332	251281	99935	25377	11325
1991	104698000	66150600	43822800	33308900	12986200	7045160	4547540	3301080	992154	506146	253726	180892	114019	60732	29987	18965
1992	102065000	66407100	45638100	34923300	24642900	13254100	8514210	6119570	2515780	1236560	564303	392870	156064	79818	35915	21450
1993	134166000	88504400	64706500	49446800	26254100	12018300	6776970	4595860	4737410	1827870	539535	331059	327549	13513 9	37704	17477
1994	117966000	75220300	54663500	42590400	37721500	20635000	13711300	1020390 0	3608010	1840450	901904	647590	279799	14722 0	70332	44079
1995	34495100	21844200	15723200	12411700	32795600	15052200	8539350	6030500	8153380	3210520	996184	628035	562716	23686 3	69032	33376
1996	63289600	39840000	26697800	21153300	9704990	4739910	2776040	2034090	4929130	2116750	741105	496862	552825	25612 2	86558	46235
1997	54156600	35398700	24069900	19267000	16673100	9178160	5952580	4503380	1687730	851918	382553	277420	466514	25072 1	112774	69691
1998	117042000	77328300	50957400	39603800	15080300	7040770	3914610	2718180	3677240	1454900	412922	255671	295099	12625 4	34237	15834
1999	74500100	49617400	35082100	27401300	30648800	17354600	11502600	8416790	2179210	1135260	530294	370824	226553	12244 1	55951	33617
2000	69854200	47175000	32734400	25712700	20799400	10177900	5814700	4096690	6590060	2735510	848757	529380	327287	14506 7	42242	20216
2001	60117900	40506900	28529200	22322700	20152300	9958470	5643940	3944530	3327360	1391530	432501	269589	463211	20888 0	60928	29277
2002	78168200	52198200	36002700	28363200	17183300	8075610	4385090	3057270	3145570	1225070	338622	208100	247810	10717 1	27504	12612
2003	107371000	72779400	51750100	40187000	22001100	11413400	6716390	4769940	2472400	1108010	389381	251041	183741	89688	29516	15146
2004	179615000	12108100 0	87644500	68395600	29711800	12935500	6652790	4485680	3681550	1291670	300359	170923	210440	81472	17136	6877

Year/Age Quarter	A0_S1	A0_S2	A0_S3	A0_S4	A1_S1	A1_S2	A1_S3	A1_S4	A2_S1	A2_S2	A2_S3	A2_S4	A3+_S1	A3+_S2	A3+_S3	A3+_S4
2005	61291800	40898100	29573900	23515100	50922700	25516300	14991000	1095090 0	3485310	1518120	534353	350486	141127	64333	21183	11016
2006	82106800	52195000	37304000	29936600	17472700	7829080	4364930	3106770	8534280	3220890	951063	593812	292053	11646 7	31795	14988
2007	63717700	39737300	27939700	22601100	22580400	10244200	5770680	4237000	2441430	939818	296274	193175	496950	20133 7	59022	29648
2008	140669000	87568900	62854000	51439600	17357000	8362940	5128100	3853360	3395740	1447050	559239	381423	185561	82338	30432	16887
2009	106893000	67757600	48722500	40169400	40007800	21569500	14378000	1129020 0	3081070	1553200	793872	589549	327525	17012 3	85100	55655
2010	101315000	61384600	42879900	34712300	31409400	15836000	10045100	7657950	9072720	4255120	1990290	1441060	531250	25790 1	117386	73372
2011	86374800	50476300	34692300	27746300	26587200	13323800	8442250	6435490	6123160	2907440	1447620	1059970	1236200	60506 3	292741	18932 8
2012	71992700	41415600	28524000	22791400	20685600	8925520	4976070	3630220	5003620	1905610	644603	432381	990629	40573 9	129592	68844
2013	187461000	11565600 0	84199300	69098300	17254100	9017270	5889160	4568030	2892030	1415910	701232	520608	411602	21732 1	104582	67942
2014	207076000	13372600 0	98296600	79595800	53385000	30531200	21087000	1661120 0	3640360	1990710	1123790	863891	480421	28493 8	157355	10949 5
2015	122969000	79254100	57622900	45267900	61495400	30613300	18245300	1331470 0	1322020 0	5920960	2333960	1572850	793495	38316 7	142707	78582
2016	185692000	11709800 0	82097100	61806200	34718200	13326800	6121320	3957330	1042150 0	3202610	643893	347692	1309490	43988 7	79177	28964
2017	238462000	15074100 0	10662000 0	82311200	46556700	22923600	13019700	9342160	3038110	1345950	516960	340258	294909	13605 0	49242	26364
2018	224072000	14611500 0	10404600 0	79987000	62500400	32076500	18511000	1324180 0	7220110	3347150	1334680	870198	288683	13913 7	52405	28021
2019	201811000	13177100 0	92451000	70787200	59971200	30678200	17377900	1354300 0	1008830 0	4641080	1813310	1427830	697438	33378 8	123070	96907

Year/Age Quarter	A0_S1	A0_S2	A0_S3	A0_S4	A1_S1	A1_S2	A1_S3	A1_S4	A2_S1	A2_S2	A2_S3	A2_S4	A3+_S1	A3+_S2	A3+_S3	A3+_S4
2020	144776000	92509000	62833900	46766300	51884100	21804100	10226600	7721410	1008990 0	3498630	826983	630251	1158150	42883 0	92061	70160
2021	155612000	99396400	67569400	50290800	33442500	14612400	7095450	5357270	5612500	2061390	535878	408397	519223	20239 1	48218	36748
2022	179246000	11469000 0	78641600	58531700	35962900	19913000	12027300	9080970	3894070	2033910	994978	758281	329991	17661 0	83363	63532
2023	94995400	60524200	40623600	30235400	41856000	12895700	4538280	3426530	6600730	1442120	147520	112426	609220	14876 8	12927	9852
2024	285066000	18239400 0	12504800 0	93071300	21621400	11926100	7177920	5419530	2490660	1293500	626335	477335	90646	48265	22532	17171
2025	0				66555300				3939320				366583			

Table 11.6.4. North Sea & 3.a Sprat. Assessment output: Estimated recruitment, spawning-stock biomass (SSB), average fishing mortality (F), and landings weight (Yield). All estimates refer to the model year, e.g., 2024 = July 2024–June 2025.

Year	Recruitment	High	Low	SSB	High	Low	Catches	F ages 1-2	High	Low
	(thousands)			(tonnes)			(tonnes)	(per year)		
1974	395307000	703815986	222029092	550749	876910	345901	463344	1.152	1.8	0.738
1975	515815000	904875514	294035047	548638	886462	339556	732312	1.771	2.596	1.208
1976	231803000	403378993	133206319	437460	708945	269938	628598	1.856	2.649	1.301
1977	379838000	642884976	224421027	305567	475060	196546	385257	1.807	2.562	1.274
1978	626428000	1142270916	343536751	319684	500536	204177	458804	1.231	1.972	0.768
1979	351927000	627055298	197514659	525867	871254	317400	463638	0.772	1.429	0.417
1980	206302000	327258001	130051871	406004	688292	239490	387434	2.396	3.283	1.749
1981	61178700	93245256	40139665	273370	404180	184896	280582	1.115	1.697	0.733
1982	37138900	53065794	25992222	173414	258070	116528	162357	0.917	1.361	0.617
1983	50514300	69983798	36461218	89860	122132	66116	115440	1.563	1.987	1.229
1984	24950900	35744380	17416651	68584	90100	52206	113444	0.806	1.21	0.536
1985	16828500	23571466	12014459	69002	91777	51879	62514	1.178	1.602	0.867
1986	60994900	85710685	43406231	25630	34048	19293	27520	0.975	1.407	0.676
1987	35362100	48943223	25549566	57800	78315	42659	53942	0.317	0.498	0.201
1988	50723500	72062392	35703415	62820	81739	48279	103652	1.136	1.493	0.865
1989	42693300	59068654	30857617	48548	64566	36503	58420	0.295	0.537	0.162
1990	61190700	83337152	44929563	46552	61554	35206	78180	1.41	1.827	1.088
1991	104698000	138446544	79176200	91196	119766	69442	125815	0.611	0.939	0.398
1992	102065000	135429634	76920124	129299	164708	101502	156471	0.778	1.104	0.549
1993	134166000	194210183	92685745	172182	219003	135371	208848	1.394	1.749	1.111
1994	117966000	158471747	87813616	136518	185747	100336	424206	0.68	0.94	0.492
1995	34495100	46654743	25504629	199490	261741	152045	446555	1.293	1.652	1.012
1996	63289600	84806110	47232133	113378	143695	89458	95496	1.102	1.452	0.837
1997	54156600	72219244	40611576	116169	148233	91041	125174	0.79	1.101	0.567
1998	117042000	158083724	86655535	144523	182614	114377	188907	1.446	1.785	1.172
1999	74500100	99919838	55547177	146499	191680	111967	243158	0.777	1.096	0.55
2000	69854200	93773541	52036099	194726	249571	151934	222027	1.317	1.709	1.015
2001	60117900	79715103	45338484	130813	168148	101768	153321	1.306	1.72	0.992
2002	78168200	105366838	57990423	118733	151117	93289	174713	1.452	1.813	1.162
2003	107371000	144262493	79913576	147627	190678	114296	174988	1.168	1.535	0.888
2004	179615000	238001316	135551974	188418	244006	145494	231352	1.749	2.143	1.428
2005	61291800	79609722	47188769	243432	314639	188340	280275	1.161	1.501	0.897
2006	82106800	106663124	63203911	180827	227927	143460	78028	1.371	1.728	1.088
2007	63717700	82496943	49213282	144673	181928	115047	99902	1.252	1.608	0.975
2008	140669000	180318237	109738027	109411	136324	87811	69892	0.997	1.374	0.723
2009	106893000	137801916	82916942	194409	242876	155614	170934	0.621	0.901	0.428

Year	Recruitment	High	Low	SSB	High	Low	Catches	F ages 1-2	High	Low
	(thousands)			(tonnes)			(tonnes)	(per year)		
2010	101315000	133535210	76869084	186216	229206	151289	145415	0.714	1.011	0.504
2011	86374800	114018023	65433568	162235	200644	131179	122472	0.607	0.914	0.403
2012	71992700	92981083	55741971	133684	165176	108196	96030	1.106	1.421	0.861
2013	187461000	246115975	142784825	103821	129046	83527	60207	0.65	1.202	0.352
2014	207076000	273766704	156631428	227737	290156	178746	190268	0.495	0.717	0.341
2015	122969000	162791724	92887861	336762	425157	266746	298227	1.007	1.336	0.76
2016	185692000	240732844	143235623	229471	289289	182022	227169	1.925	2.267	1.635
2017	238462000	309129109	183949437	189248	239111	149783	135824	1.039	1.381	0.782
2018	224072000	298143041	168403264	226619	283549	181119	190779	1.018	1.364	0.76
2019	201811000	267232133	152405623	241888	306351	190990	137029	0.921	1.309	0.648
2020	144776000	188881417	110969573	321963	409938	252868	182205	1.523	1.909	1.216
2021	155612000	214531044	112874548	157159	196960	125401	80183	1.387	1.855	1.038
2022	179246000	241070772	133276748	165053	216458	125856	89722	0.586	0.99	0.347
2023	94995400	132711280	67998184	224679	288368	175057	92850	2.511	2.882	2.188
2024	285066000	497954831	163192762	97068	126606	74421	83829	0.599	1	0.359
2025	169486922*			274902	449480	168130				

* Geometric mean recruitment (2013–2023)

Table 11.9.1. North Sea and Division 3.a Sprat. Input to forecast (years and age refer to the model year, e.g., 2024 = July 2024–June 2025).

Age	Age 0	Age 1	Age 2	Age 3
Stock numbers(2025) (millions)	169487	66555	3939	367
Exploitation pattern S1	0.0055	0.7507	1.1161	1.0048
Exploitation pattern S2	0.0274	0.6917	2.0039	2.1671
Exploitation pattern S3	0.0000	0.0000	0.0000	0.0000
Exploitation pattern S4	0.0000	0.0000	0.0000	0.0000
Weight in the stock S1 (gram)	5.1328	8.5577	10.8297	13.2209
Weight in the catch S1 (gram)	5.1328	8.5577	10.8297	13.2209
Weight in the catch S2 (gram)	5.7648	9.6886	13.2460	14.9407
Weight in the catch S3 (gram)	5.0533	8.6866	12.1194	14.3000
Weight in the catch S4 (gram)	5.8937	9.1178	12.0813	15.3616
Proportion mature(2025)	0.0000	0.4093	0.8719	0.9468
Proportion mature(2026)	0.0000	0.4093	0.8719	0.9468
Natural mortality S1	0.4453	0.4267	0.4050	0.4050
Natural mortality S2	0.3713	0.3527	0.2760	0.2760
Natural mortality S3	0.2953	0.2810	0.2717	0.2717
Natural mortality S4	0.3353	0.3190	0.2993	0.2993

Table 11.9.2. Sprat North Sea Division 3.a. Short-term predictions options table. Years refer to the model year, e.g., 2024 = July 2024–June 2024.

Catch options. Catches and SSB are in thousands of tonnes.					
<i>3-year average weight-at-age was used to calculate SSB. Recruitment(2025) = geometric average 2014–2023.</i>					
Basis	Catches(2025)	F(2025)	SSB(2026)	SSB change*	TAC change**
F _{cap}	236114	1.01	225010	-18%	214%
F=0.0	1	0	313238	14%	-100%
F=0.1	30648	0.1	301371	10%	-59%
F=0.2	59351	0.2	290352	6%	-21%
F=0.3	86259	0.3	280110	2%	15%
F=0.4	111508	0.4	270583	-2%	48%
F=0.5	135222	0.5	261716	-5%	80%
F=0.6	157515	0.6	253456	-8%	109%
F=0.7	178489	0.7	245757	-11%	137%
F=0.8	198241	0.8	238576	-13%	163%
F=0.9	216855	0.9	231873	-16%	188%
F=1.0	234413	1	225613	-18%	211%
Bescapement with-out F _{cap}	534453	5.118	135951.6	-50.50%	609.60%

* SSB 1st July 2025 relative to SSB 1st July 2026

** Catch (July 2025-June 2026) relative to the sum of the TACs (75 321 tonnes) for July 2022–June 2023 in Subarea 4 and Division 3.a.